

Pharmacology of Paracetamol

Pongsatorn Meesawatsom BPharm, PhD
Department of Pharmacology
Faculty of Pharmacy
Mahidol University

Objectives

- ★ อธิบายกลไกการออกฤทธิ์และผลทางเภสัชวิทยาของ paracetamol และยาแก้พิษของ paracetamol ได้
- ★ อธิบาย metabolism ของยา paracetamol และความสำคัญทางคลินิกได้

Mechanisms of paracetamol in inhibition of COX isozymes

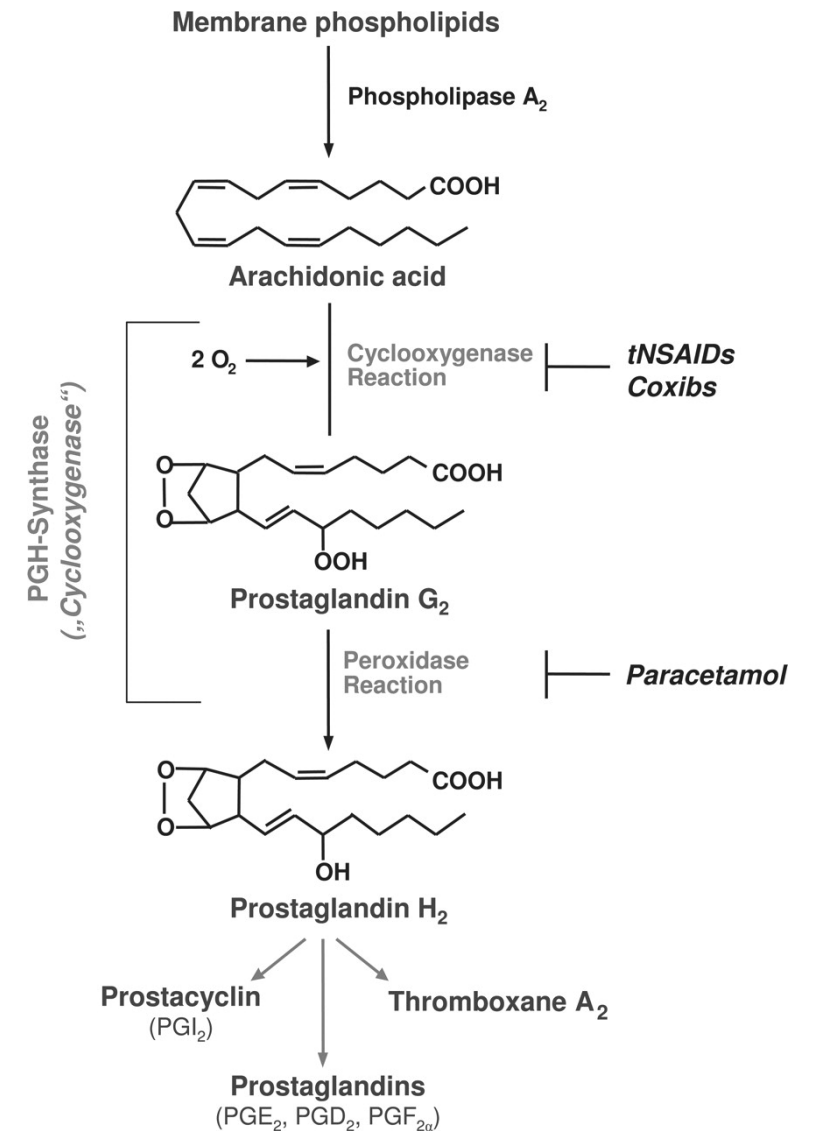
- ☀ Preferential COX-2 inhibitors

- ☾ Act as a reducing substrate in peroxidase reaction which converts PGG_2 to PGH_2
- ☾ ↓ PGE_2 synthesis

- ☀ Inhibit a different step of NSAIDs/coxibs

- ☀ Acts predominantly in the CNS

- ☾ The inhibitory effect is attenuated in high arachidonic acid concentration and oxidative stress → no anti-inflammatory effect



Ann Rheum Dis. 2012 Jan;71(1):20-5.

Pharmacol Res. 2016 Jul;109:119-31.

CNS Drug Rev. 2006 Fall-Winter;12(3-4):250-75.

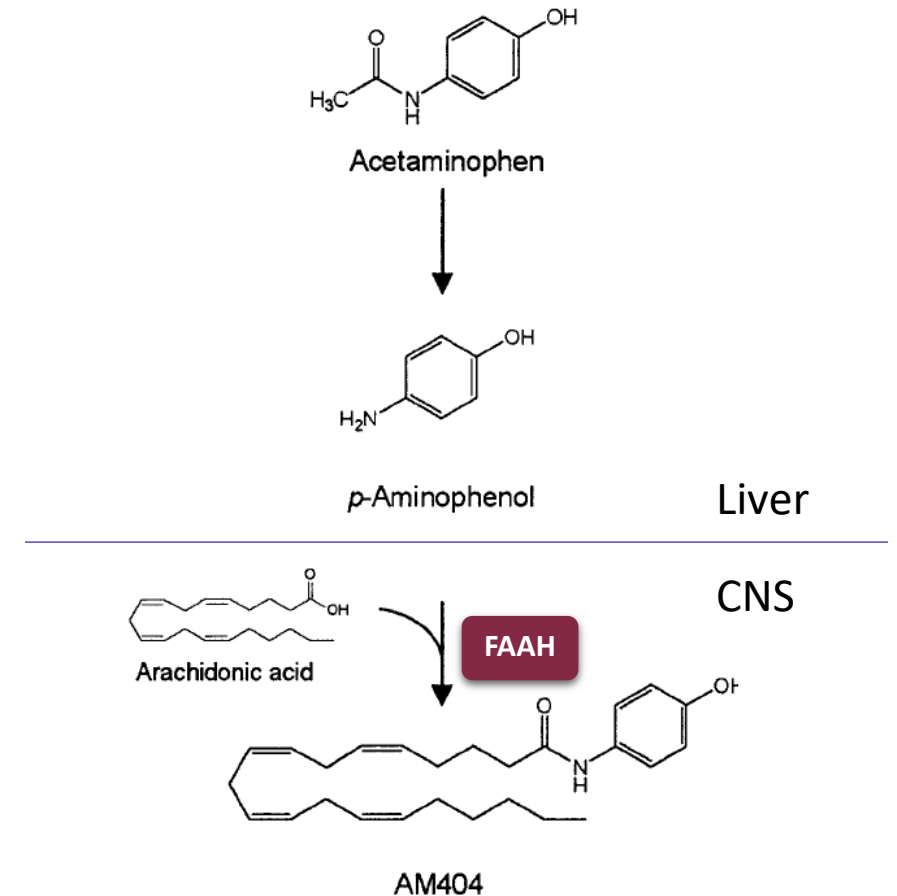
Current proposed mechanisms of actions of paracetamol and AM404

★ Paracetamol

- ☾ Reversible competitive inhibitor of cyclooxygenase (COX) inhibition (COX-2 > COX1)
- ☾ Inhibit COX-2 in CNS

★ N-arachidonoylphenolamine (AM404)

- ☾ Increase availability of an endocannabinoid anandamide
- ☾ Modulate ion channels e.g. TRPV1 and T-type calcium channel
- ☾ Increase serotonergic and opioidergic tones in descending inhibitory pain pathway



FAAH = fatty acid amide hydrolase

Paracetamol

☀ Pharmacological effects

- ☾ Analgesic for mild to moderate pain
- ☾ Normalized set point of thermoregulation center in hypothalamus → antipyretic
- ☾ Sweating

☀ Pros (when used in recommended dose)

- ☾ Relatively safe across most patient conditions (but long-term use still has risks)
- ☾ Less drug interactions

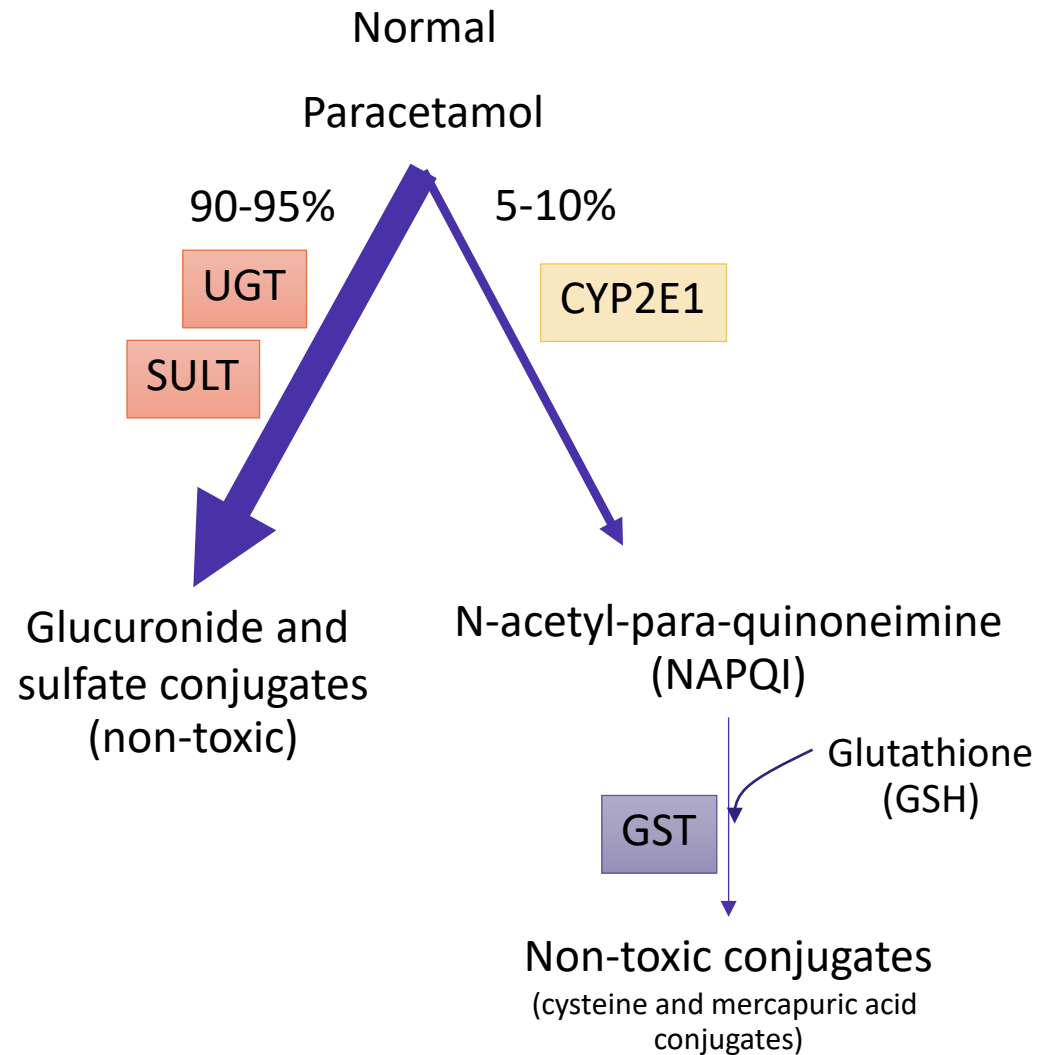
☀ Cons

- ☾ Limited analgesic effect
- ☾ No anti-inflammatory effect

Therapeutic uses

	NSAIDs	Paracetamol
Antipyretic	✓	✓
Analgesic	✓	✓
Anti-inflammatory	✓	

Paracetamol metabolism



Overdose
Glutathione depletion
CYP2E1 induction
↑NAPQI

Covalent bonding of hepatocellular protein
↓
Hepatotoxicity

UGT = UDP-glucuronosyltransferase
SULT = sulfotransferases
GST = glutathione S-transferases

Paracetamol metabolism

- ✱ The majority of acetaminophen is metabolized by the liver via glucuronidation and sulfation to nontoxic metabolites
- ✱ Less than 10% of acetaminophen is metabolized by CYP2E1, which converts acetaminophen to N-acetyl-p-benzoquinone imine (NAPQI), the toxic intermediate metabolite
- ✱ NAPQI is rapidly detoxified by hepatic glutathione and excreted in the urine

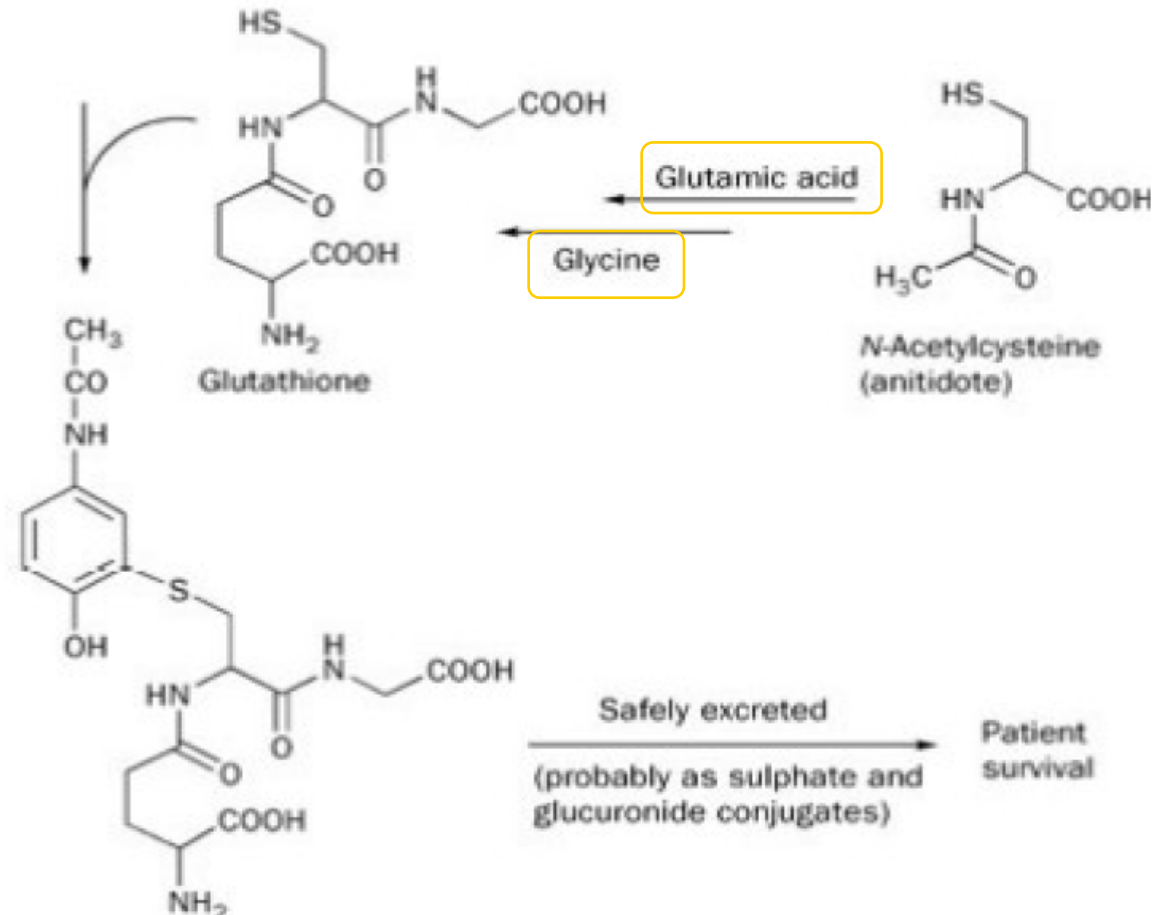
Paracetamol intoxication

- ★ Toxic effect seen after 7.5-10 g intake or 150 mg/kg
- ★ Fatal dose is single ingestion of 20-25 g
- ★ Hepatic injury from NAPQI occurs within the first 24 hr

Time post ingestion	Symptoms
12 -24 hr	N/V/D, diaphoresis, anorexia
24-48 hr	All symptom resolve** Increasing in AST, ALT, bilirubin, prothrombin time
72-96 hr	Peak hepatitis symptoms
7-8 day	Recovery

N-acetylcysteine is an antidote for paracetamol overdose

N-Acetyl-benzoquinoneimine



Lower paracetamol dose to 50-75% in patients with

- ✱ CYP2E1 overactivity

- ☾ Chronic alcoholism/ regularly alcohol intake > 3 drink/day
- ☾ Concomitant use of phenytoin, carbamazepine, rifampin or other enzyme inducers

- ✱ Malnutrition